

## Calculus I Lesson 47

$$\text{Formula A: } \int \frac{1}{\sqrt{a^2 - u^2}} du = \arcsin\left(\frac{u}{a}\right) + C$$

$$\text{Formula B: } \int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$$

$$\text{Formula C: } \int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \operatorname{arcsec}\left(\frac{|u|}{a}\right) + C$$

$$\int \frac{1}{\sqrt{a^2 - u^2}} du = \arcsin\left(\frac{u}{a}\right) + C$$

### Question 1:

Find  $\int \frac{dx}{\sqrt{25 - x^2}}$

Hint: Let  $a^2 = 25 \Rightarrow a = 5$

Let  $u^2 = x^2 \Rightarrow u = x \Rightarrow x = u;$

$$\frac{du}{dx} = 1 \Rightarrow du = dx$$

a) Write the integral  $\int \frac{1}{\sqrt{25 - x^2}} dx$  in terms of  $u$ :  $\int \underline{\hspace{1cm}} \cdot du$

b)  $\int \frac{1}{\sqrt{25 - x^2}} dx = \underline{\hspace{1cm}} ?$

$$\int \frac{1}{\sqrt{a^2 - u^2}} du = \arcsin\left(\frac{u}{a}\right) + C$$

## Question 2:

Find  $\int \frac{1}{\sqrt{1 - (x + 4)^2}} dx$

Hint: Let  $a^2 = 1 \Rightarrow a = 1$

Let  $u^2 = (x + 4)^2 \Rightarrow u = x + 4$

Hence,  $\frac{du}{dx} = 1 \Rightarrow du = dx$

a) Write the integral  $\int \frac{1}{\sqrt{1 - (x + 4)^2}} dx$  in terms of  $u$ :  $\int \underline{\hspace{1cm}} \cdot du$

b)  $\int \frac{1}{\sqrt{1 - (x + 4)^2}} dx \underline{\hspace{1cm}}$

$$\int \frac{1}{\sqrt{a^2 - u^2}} du = \arcsin\left(\frac{u}{a}\right) + C$$

### Question 3:

Find  $\int \frac{1}{\sqrt{1-25x^2}} dx$

Hint: Let  $a^2 = 1 \Rightarrow a = 1$

Let  $u^2 = 25x^2 \Rightarrow u = 5x$

Hence,  $\frac{du}{dx} = 5 \Rightarrow du = 5dx \Rightarrow \frac{1}{5} du = dx$

a) Write the integral  $\int \frac{1}{\sqrt{1-25x^2}} dx$  in terms of  $u$ :  $\int \frac{1}{\sqrt{1-u^2}} \cdot \frac{1}{5} \cdot du = \frac{1}{5} \cdot \int \underline{\hspace{1cm}} \cdot du$

b)  $\int \frac{1}{\sqrt{1-25x^2}} dx = \underline{\hspace{1cm}}$

### Question 4:

$$\int \frac{1}{\sqrt{a^2 - u^2}} du = \arcsin\left(\frac{u}{a}\right) + C$$

Find  $\int \frac{7}{12\sqrt{8-14x^2}} dx$ .

Hint: Let  $a^2 = 8 \Rightarrow a = \sqrt{8}$

Let  $u^2 = 14x^2 \Rightarrow u = \sqrt{14} \cdot x$

Hence,  $\frac{du}{dx} = \sqrt{14} \Rightarrow du = \sqrt{14} dx \Rightarrow \frac{1}{\sqrt{14}} du = dx$

a) Write the integral  $\int \frac{7}{12\sqrt{8-14x^2}} dx$  in terms of  $u$ :  $\int \frac{7}{12\sqrt{8-14x^2}} dx = \frac{7}{12} \cdot \int \frac{1}{\sqrt{8-14x^2}} \cdot dx = \underline{\quad? \quad} \cdot du$

b)  $\int \frac{7}{12\sqrt{8-14x^2}} dx = \underline{\quad? \quad}$

### Question 5:

$$\int \frac{1}{\sqrt{a^2 - u^2}} du = \arcsin\left(\frac{u}{a}\right) + C$$

Find  $\int \frac{12}{5\sqrt{7-21x^2}} dx$ .

Hint: Let  $a^2 = 7 \Rightarrow a = \sqrt{7}$

Let  $u^2 = 21x^2 \Rightarrow u = \sqrt{21} \cdot x$

Hence,  $\frac{du}{dx} = \sqrt{21} \Rightarrow du = \sqrt{21} dx \Rightarrow \frac{1}{\sqrt{21}} du = dx$

a) Write the integral  $\int \frac{12}{5\sqrt{7-21x^2}} dx$  in terms of  $u$ :  $\int \frac{12}{5\sqrt{7-21x^2}} dx = \frac{12}{5} \cdot \int \frac{1}{\sqrt{7-21x^2}} \cdot dx = \underline{\quad? \quad} \cdot du$

b)  $\int \frac{12}{5\sqrt{7-21x^2}} dx = \underline{\quad? \quad}$